Revisit for AG,ANT and RT

April 23, 2026

Chapter 1

Algebraic Geometry

Let us review concepts from Algebraic Geometry.
1. Let $A \in \mathrm{Ring}$, then we have $\mathrm{Sch}_{/\mathrm{Spec}(A)} \simeq \mathrm{Ring}_{A/}$. The correspondence is given by $(X \mapsto \mathrm{Spec}(A)) \rightsquigarrow A \to \Gamma(X, \mathcal{O}_X)$.
2. If we consider $X \mapsto \mathbb{A}^1_A$, it corresponds to $A[t] \mapsto \Gamma(X, \mathcal{O}_X)$ which uniquely determined by the image of t. In this sense, we obtain that $(X \mapsto \mathbb{A}^1_A) \rightsquigarrow \Gamma(X, \mathcal{O}_X)$. Moreover, we have $\mathrm{Hom}_S(X, \mathbb{A}^1_S) \simeq \mathrm{Hom}_{\mathrm{Spec}(\mathbb{Z})}(X, \mathbb{A}^1) \simeq \Gamma(X, \mathcal{O}_X)$.
3. Let A be a ring, we have $\mathrm{Proj}_{/\mathrm{Spec}(A)} \simeq \mathrm{Mod}_A$.
4. Structure sheaf $O_X : X^{\mathrm{op}} \to \mathrm{Ring}$.
5. Let $j_!$ denotes extension by zero and U is an open subscheme of X then we obtain that $\Gamma(U, \mathcal{F}) \cong \mathrm{Hom}_{\mathcal{O}_X}(j_! \mathcal{O}_U, \mathcal{F})$.
6. Serre's criterion for Normality: A Noetherian ring A is normal if and only if it has properties R1 and S2. R1 means $A_{\mathfrak{p}}$ is regular(smooth) for any codimension one prime ideal $\mathfrak{p}$. S2 means $\mathrm{depth}(A_{\mathfrak{p}}) = \min(2, \dim(A_{\mathfrak{p}}))$ for any prime ideal.
7. $\mathrm{depth}(A_{\mathfrak{p}})$ means the maximal length of regular sequence $(a_1, a_2, \ldots,)$ in $\mathfrak{p}$. Regular means that a_i is not a zero divsor in $A/(a_1, \ldots, a_i)$.
8. Sheaf of O_X-module:

$$\begin{tikzcd} & \mathcal{M}od \arrow[d, "P"] \\ \mathrm{Open}(X)^{\mathrm{op}} \arrow[ur, "\mathcal{F}"] \arrow[r, "\mathcal{O}_X"'] & \mathrm{Ring} \end{tikzcd}$$

9. $f : X \to Y$, then $f^*\mathscr{G} = f^{-1}\mathscr{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$
10. Let M be A-module and $X = \mathrm{Spec}(A)$. Then $\widetilde{M} := \underline{M} \otimes \mathcal{O}_X$ where $\underline{M}$ denotes the constant sheaf of M.
11. Ideal sheaf is the kernal of $i^\# : \mathcal{O}_X \to i_*\mathcal{O}_Y$ induced by closed immersion $i : Y \to X$.
12. We call a sheaf $\mathscr{F}$ is quasi-coherent if for any affine open subscheme U, there exist an $\mathcal{O}_X(U)$-module M such that $\mathscr{F}|_U \simeq \widetilde{M}$. Hence, we call it coherent if X is Noetherian.
13. Let X be any affine scheme and $\mathscr{F}$ be an arbitrary quasi-coherent sheaf. Then we obtain that $H^1(X, \mathscr{F}) = 0$. It means that if we consider the exact sequence of ideal sheaf

$$0 \to \mathcal{I}_Y \to \mathcal{O}_X \to i_*\mathcal{O}_Y \to 0$$

The global section functor is exact

$$0 \to \Gamma(X, \mathcal{I}_Y) \to \Gamma(X, \mathcal{O}_X) \to \Gamma(X, i_*\mathcal{O}_Y) \to H^1(X, \mathcal{I}_Y) = 0.$$

Equivalently, we say any regular function on closed subscheme of X can be extended to X.

14. For any scheme X, S2 condition is equivalent to $j_*(\mathcal{O}_U) \simeq \mathcal{O}_X$ for any $\operatorname{codim}(X \setminus U) \geq 2$
15. For open subscheme U of X, there is an exact sequence

$$0 \to i_!\mathcal{O}_U \to \mathcal{O}_X \to j_*\mathcal{O}_Z \to 0$$

where $i_!\mathcal{O}_U$ is the push-forward with compact support. The inclusion is due to the extension by zero.
16. We also have exact sequence

$$0 \to \mathcal{H}^0_Z(\mathcal{O}_X) \to \mathcal{O}_X \to i_*\mathcal{O}_U \to \mathcal{H}^1_Z(\mathcal{O}_X) \to \dots$$

where $\mathcal{O}_X \to i_*\mathcal{O}_U$ is restriction. $\mathcal{H}^0_Z(\mathcal{O}_X)$ means the sheaf of supporting on Z. It also can be defined by the sheafication of presheaf functor $U \mapsto \Gamma_Z(X, \mathcal{O}_X) \cap U$ where Γ_Z means supporting on Z.
17. In the projective case, we have an similar exact sequence. Given that an projecitve shceme $X = \operatorname{Proj}(S)$, then we interpret it as the open subscheme of CX excluding zero. The exact sequence follows

$$0 \to H^0_{\mathfrak{m}}(S) \to \Gamma(CX, \mathcal{O}_{CX}) = S \to \Gamma_*(\mathcal{O}_X) \to H^1_{\mathfrak{m}}(S) \to H^1(CX, \mathcal{O}_{CX}) = 0.$$

18. Let M be R-module, we define $\operatorname{Ass}(M) = \{\operatorname{Ann}_R(m)\}_{m \in M} \cap \operatorname{Spec}(R)$. Then $\operatorname{Supp}(M) = \bigcup_{\mathfrak{p} \in \operatorname{Ass}(M)} V(\mathfrak{p})$. To see that we consider if $\mathfrak{p} = \operatorname{Ann}_R(m)$ then $M_{\mathfrak{p}} \neq 0$.
19. Let $X = \operatorname{Spec}(R)$ and $f \in R$, then $\operatorname{Supp}(f) = V(\operatorname{Ann}_R(f))$. Hence, if f is not zero divsor, then $\operatorname{Supp}(f) = X$.
20. If we consider ${}_RR$, then $\operatorname{Ass}(R) = \{\sqrt{\mathfrak{q}_1}, \dots, \sqrt{\mathfrak{q}_n}\}$ where $(0) = \mathfrak{q}_1 \cap \dots \cap \mathfrak{q}_n$ be the primary decomposition of zero ideal.
21. Suppse that I is radical and $Z = V(I)$. We have $\mathcal{H}^0_Z(\mathcal{O}_X) = \{x \in \mathcal{O}_X \mid \sqrt{\operatorname{Ann}_R(x)} \supseteq I\}$. Hence, we can obtain that $\mathcal{H}^0_Z(\mathcal{O}_X) = \bigcap_{\mathfrak{p}_i \not\supseteq I} \mathfrak{q}_i$.
22. For a graded ring S with M is a graded S-module. Define $\widetilde{M}(D_+(f)) := \{m/f^n | \deg(m) = n\}$.
23. Let S be a graded ring, $S(n)$ denotes twisted module defined by $S(n)_d = S_{n+d}$. Define the twisted sheaf $\mathcal{O}_X(n) := \widetilde{S(n)}$ and twisted sheaf of module $\mathscr{F}(n) := \mathscr{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X(n)$.
24. Let S be the graded ring generated by S_1 as S_0-algebra. Let $X = \operatorname{Proj}(S)$. Then

- $\mathcal{O}_X(n)$ is an invertible sheaf on X;
- $\widetilde{M}(n) \simeq \widetilde{M(n)}$;
- $\mathcal{O}_X(n) \otimes \mathcal{O}_X(m) \simeq \mathcal{O}_X(n+m)$

25. The section functor is given by $\Gamma_*(\mathscr{F}) = \bigoplus_{n \in \mathbb{Z}} \Gamma(X, \mathscr{F}(n))$. It is right adjoint functor of $\widetilde{(\cdot)}$. Hence if S is a graded ring generated by S_0-algebra S_1 and $\mathscr{F}$ is quasi-coherent, then $\Gamma_*(\mathscr{F}) \simeq \mathscr{F}$.
26. Extension of Sections: Let X be a scheme and $\mathscr{L} \in \operatorname{Pic}(X), \mathscr{F} \in \operatorname{Qcoh}(X)$. Suppose X is quasi-compact and has a finite open cover $\{U_i\}$ such that $\mathscr{L}|_{U_i}$ is free and $U_i \cap U_j$ is quasi compact. Given that $t \in \Gamma(X_f, \mathscr{F})$, then for some $n > 0$, the section $f^n t \in \Gamma(X_f, \mathscr{F} \otimes \mathscr{L}^{\otimes n})$ can be extended to a global section of $\mathscr{F} \otimes \mathscr{L}^{\otimes n}$.
27. we call a morphism $f : X \to Y$ is projective if f can be factorized as form $X \to \mathbb{P}^n_Y \to Y$ where $X \to \mathbb{P}^n_Y$ is a closed immersion. In this case, we call X is a projective scheme over Y. If $Y = \operatorname{Spec}(A)$, then exists a graded ring S finitely generated by $A = S_0$-algebra S_1.
28. we call a morphism is quasi projective if $X \to \mathbb{P}^n_Y$ is a locally closed immersion which means there exists closed subscheme Z such that $X \to Z$ is an open immersion.

29. For any scheme Y, define twisting sheaf $\mathcal{O}(d)$ on $\mathbb{P}^r_Y$ by $g^*(\mathcal{O}(d))$ where $g : \mathbb{P}^r_Y \to \mathbb{P}^r_{\mathbb{Z}}$.
30. X is a shceme over Y. We call invertible sheaf $\mathscr{L}$ is very ample relative to Y if there is a locally closed immersion $i : X \to \mathbb{P}^r_Y$ for some r such that $i^*(\mathcal{O}(1)) \simeq \mathscr{L}$.
31. We call $\mathscr{F}$ is generated by global sections if there is a family of global sections $\{s_i\}$ such that $\mathscr{F}_x$ is generated by $\{s_i(x)\}$.
32. Let A be a finitely generated k-algebra with graded structure. Let X be projecitve over $\mathrm{Spec}(A)$ and $\mathscr{F}$ is an arbitrary coherent sheaf. Then $\Gamma(X, \mathscr{F})$ is a finitely generated A-module.
33. Let Y is quasi-projecitve over $\mathrm{Spec}(A)$. Equivalently, Y is a quasi-projecitve subscheme of $\mathbb{P}^n_A$. Then we have a very ample line bundle of Y by pull-back $i^*(\mathcal{O}_{\mathbb{P}^n_A}(1))$.

Chapter 2

Algebraic Number Theory

First let we consider some basic extensions:

Type A Quadratic Extensions $[L : K] = 2$;

Type B Cyclotomic Extensions $K(\zeta_n)/K$;

Type C Kummer Extensions $L = K(\sqrt[n]{a})$;

Type D Pure Cubic Extensions $L = K(\sqrt[3]{a})$;

Type E Cyclic Cubic Extensions $\mathrm{Gal}(L/K) \cong C_3$;

Type F Biquadratic Extensions $L = K(\sqrt{a}, \sqrt{b})$;

Type G Quaternion Extensions $\mathrm{Gal}(L/K) \cong Q_8$;

Type H Dihedral Extensions $\mathrm{Gal}(L/K) \cong D_{2n}$;

Type I Complex Multiplication Fields $L = \mathbb{Q}(\zeta_n + \zeta_n^{-1})$;

Type J Eisenstein Extension $L = \mathbb{Q}(\omega)$ where ω is a root of an Eisenstein polynomial.

Type K Artin-Schreier Extensions $L = K(y)$ with $y^p - y = a \in K$.

For Type C, the base field K must contains the roots of unity $\mu_n = \langle \zeta_n \rangle$. For Type K, $\mathrm{char}(K) = p$.

Let us review concepts from Algebraic Number Theory.

1. Let L/K be a finite seperable extension. The trace operator Tr is a bilinear form on K-vector space L.

2. The trace ideal $\mathfrak{T}_{L/K}$ is defined by scale ideal $\mathfrak{s}(\mathcal{O}_L) = \langle \mathrm{Tr}_{L/K}(xy) \mid x, y \in \mathcal{O}_L \rangle_{\mathcal{O}_K}$. The extension is tamely ramified if and only if $\mathfrak{T}_{L/F} = \mathcal{O}_K$.

3. The discriminant of field L is given by $\mathrm{dics}(\langle \alpha_1, \ldots, \alpha_n \rangle) = \det(\mathrm{Tr}(\alpha_i \alpha_j))$ where $\{\alpha_i\}$ is an integral basis.

4. We define the Different ideal $\mathfrak{D}_{L/K}$ by the inverse of dual lattice $(\mathcal{O}_L^{\#})^{-1}$.

5.